

An exact obstruction to extracting the triple term linearly

Exact certificate; independently verified by a second agent

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For a concatenation of m independently specified palindrome blocks $\rho_i \text{rev}(\rho_i)$ on n labels, write $R(\sigma)$ for the ratio of the probability of target order σ to $1/n!$. Let $S(\sigma)$ be the triple contribution defined in the companion palindrome notes. Both $R - 1$ and S are vectors indexed by S_n .

Proposition 1. *Already for $n = 5$ and $m = 3$, there is no linear map T on target-order vectors such that $S = T(R - 1)$ for every choice of the block permutations. This remains false if all lower-dimensional ordering marginals are included in the input, since they are linear averages of the full vector.*

The conclusion concerns an exact universal linear extraction. It does not exclude an approximate extraction, a nonlinear inequality, or a lower bound for the complete deterministic ordering error. In this example products of two block defects have degree at least six and therefore do not occur: the obstruction is cancellation between single-block contributions of lengths three, four, and five.

Proof by an exact integer certificate. For a distinct-letter pattern p of length $k \in \{3, 4, 5\}$ put

$$D_\rho(p) = N_\rho(p) - \frac{2^k}{k!},$$

where $N_\rho(p)$ is its subsequence count in the ten-letter palindrome $\rho \text{rev}(\rho)$. Let slots be $i = 0, 1, 2$ and write $j = 2 - i$. Define integer columns, indexed by full patterns $p \in S_5$,

$$A_{i,\rho}(p) = 15 \sum_{\substack{a,b \geq 0 \\ a+b \leq 2}} \frac{(2i)^a (2j)^b}{a!b!} D_\rho(p_{a+1} \cdots p_{5-b}),$$

$$B_{i,\rho}(p) = 15 \sum_{\substack{a,b \geq 0 \\ a+b=2}} \frac{(2i)^a (2j)^b}{a!b!} D_\rho(p_{a+1} \cdots p_{5-b}).$$

The reduced-signature product expansion gives exactly

$$R - 1 = \frac{120}{15 \cdot 6^5} \sum_{i=0}^2 A_{i,\rho_i}, \quad S = \frac{120}{15 \cdot 6^5} \sum_{i=0}^2 B_{i,\rho_i}.$$

Indeed, a palindrome block agrees with $\exp(2 \sum X_i)$ through degree two; products of two defects vanish in degree five. The formula for B retains precisely its degree-three part.

Here is a signed integer array $v_{i,\rho}$, with unlisted entries zero:

i	ρ	v	i	ρ	v
0	01234	-5	0	01324	5
0	10234	-4	0	10324	4
0	23401	8	0	32401	-8
0	40213	-8	0	40312	8
0	41203	-8	0	41302	8
1	01234	2	1	01324	-2
2	01234	-1	2	01324	1

Exact subsequence counting verifies

$$\sum_{\rho} v_{i,\rho} = 0 \quad (i = 0, 1, 2), \quad \sum_{i,\rho} v_{i,\rho} A_{i,\rho} = 0,$$

whereas

$$\left(\sum_{i,\rho} v_{i,\rho} B_{i,\rho} \right) (01234) = -120.$$

All 120 coordinates of the first vector identity and of the nonzero second vector are recorded in `linear_extraction_counterexample.json`. Two independent exact programs verified them: one uses subsequence dynamic programming, the other enumerates position subsets of each palindrome directly.

To deduce the stated impossibility for actual block choices, independently sample each slot from the rational probability distribution

$$q_{i,\rho} = \frac{1}{120} + \frac{v_{i,\rho}}{960}.$$

These probabilities are nonnegative because $|v| \leq 8$, and each slot sums to one. Under the unperturbed uniform distributions, both displayed defect vectors have mean zero, by uniform relabeling. The exact identities therefore give

$$\mathbb{E}(R - 1) = 0, \quad \mathbb{E}S(01234) = -\frac{1}{6^5} \neq 0.$$

A universal linear map T would imply $\mathbb{E}S = T(\mathbb{E}(R - 1)) = 0$, a contradiction. $\square$

Reproduction. The discovery script is `find_extraction_counterexample.py`; the matrix definition is `check_linear_extraction.py`. The independent standard-library verifier is `../independent_directions/review_linear_extraction_counterexample.py`. The exact identities, not the numerical ranks used during discovery, are the certificate.